



Application Note

MI Data - L11-5v HIQ Scripts

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Document Change History

Date	Section(s)	Description	Author
November 6, 2019	All	Rev A: Initial document	PJ

1. Disclaimer

The MI data provided below are for reference only. It is the user's responsibility to determine whether to perform additional measurements depending on his or her use of the information.

The measurements were taken using one transducer on one Vantage System. Verasonics did not evaluate variability between different transducers nor between different Vantage Systems, so the MI values may be different if the measurements were conducted with a different transducer and/or Vantage System. Moreover, measurement errors may be introduced when using a different hydrophone or measurement system.

The measurement system and measurement procedure are developed by Verasonics following NEMA and IEC standards.

1. AIUM NEMA UD2 Acoustic output measurement standard for diagnostic ultrasound equipment. 2007.
2. IEC 62359 Ultrasonics - Field characterization - Test methods for the determination of thermal and mechanical indices related to medical diagnostic ultrasonic fields. Ed. 2.0. 2010.

3. AIUM NEMA UD3 Standard for real time display of thermal and mechanical acoustic output indices on diagnostic ultrasound equipment. 2004 – Revision 2.

2. Measurement Information

L11-5v HIQ script Mechanical Index (MI) values were measured with

Transducer: S/N UA-190015
 System: S/N KV22446
 Hydrophone: S/N D1604-117 (Precision Acoustics, Calibrated)

Note:

1. TxVoltage in the data listed below is the transmit voltage entered in the GUI.
2. Location is the position in acoustic field where the MI data was measured, where Z is the depth from transducer surface, XZ is the imaging plane and YZ is the elevation plane. (0,0,0) is the location of geometry center of a transducer.

3. MI Data

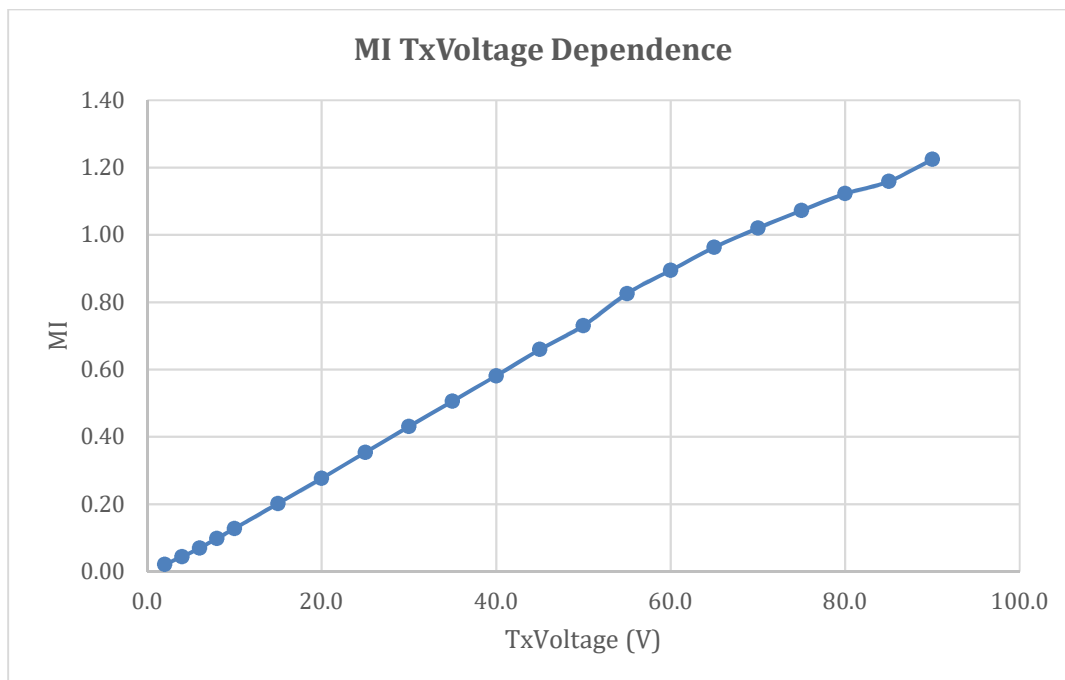
3.1 L11-5v WideBeamSC

Software release: 4.1.100-1812101200

Location: (0.00, 0.00, 12.36) mm

TxVoltage (V)	Pp (Mpa)	Pr (Mpa)	MI
2.0	0.07	0.08	0.02
4.0	0.15	0.16	0.04
6.0	0.24	0.24	0.07
8.0	0.33	0.34	0.10
10.0	0.42	0.44	0.13
15.0	0.66	0.70	0.20
20.0	0.92	0.95	0.28
25.0	1.19	1.21	0.35
30.0	1.45	1.47	0.43
35.0	1.71	1.72	0.51
40.0	2.03	1.97	0.58
45.0	2.46	2.22	0.66

50.0	2.90	2.44	0.73
55.0	3.38	2.74	0.83
60.0	3.89	2.95	0.89
65.0	4.25	3.16	0.96
70.0	4.39	3.33	1.02
75.0	4.48	3.48	1.07
80.0	4.65	3.63	1.12
85.0	4.20	3.73	1.16
90.0	5.35	3.93	1.22



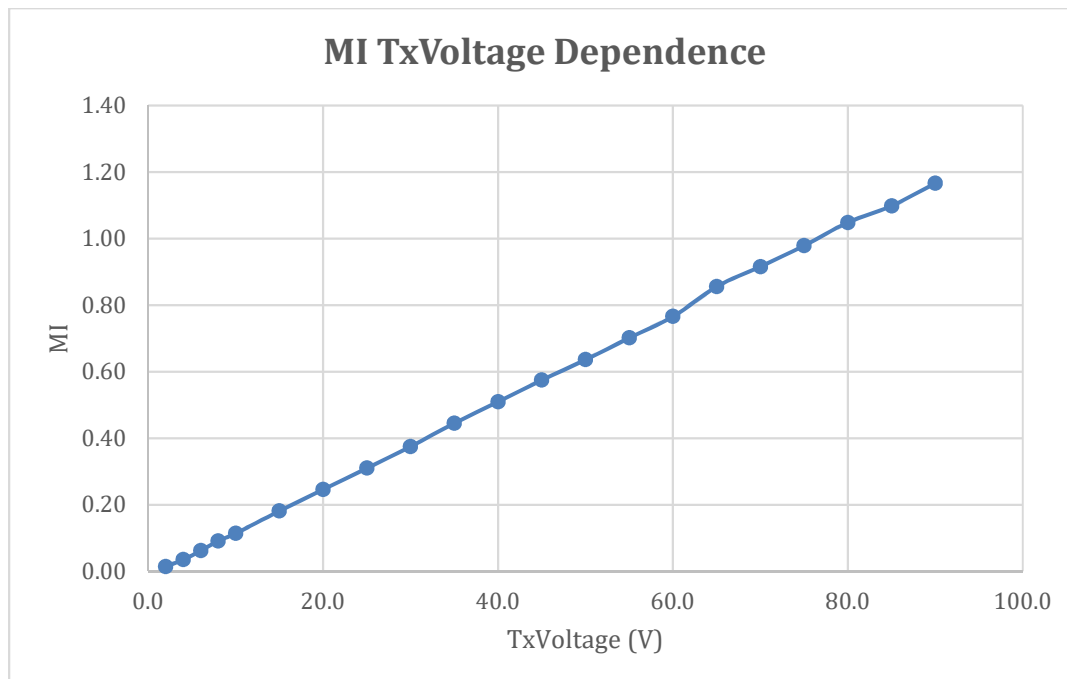
3.2 L11-5v WideBeamHISC (Inversed transmit)

Software release: 4.1.100-1812101200

Location: (0.00, 0.00, 10.98) mm

TxVoltage (V)	Pp (Mpa)	Pr (Mpa)	MI
2.0	0.04	0.04	0.01
4.0	0.11	0.10	0.04
6.0	0.18	0.17	0.06
8.0	0.26	0.25	0.09

10.0	0.34	0.32	0.11
15.0	0.54	0.50	0.18
20.0	0.74	0.68	0.25
25.0	0.94	0.85	0.31
30.0	1.14	1.03	0.38
35.0	1.34	1.22	0.44
40.0	1.55	1.39	0.51
45.0	1.76	1.57	0.57
50.0	1.96	1.73	0.64
55.0	2.17	1.91	0.70
60.0	2.37	2.08	0.77
65.0	2.57	2.32	0.86
70.0	2.78	2.48	0.92
75.0	3.03	2.65	0.98
80.0	3.37	2.83	1.05
85.0	3.71	2.96	1.10
90.0	4.17	3.14	1.17

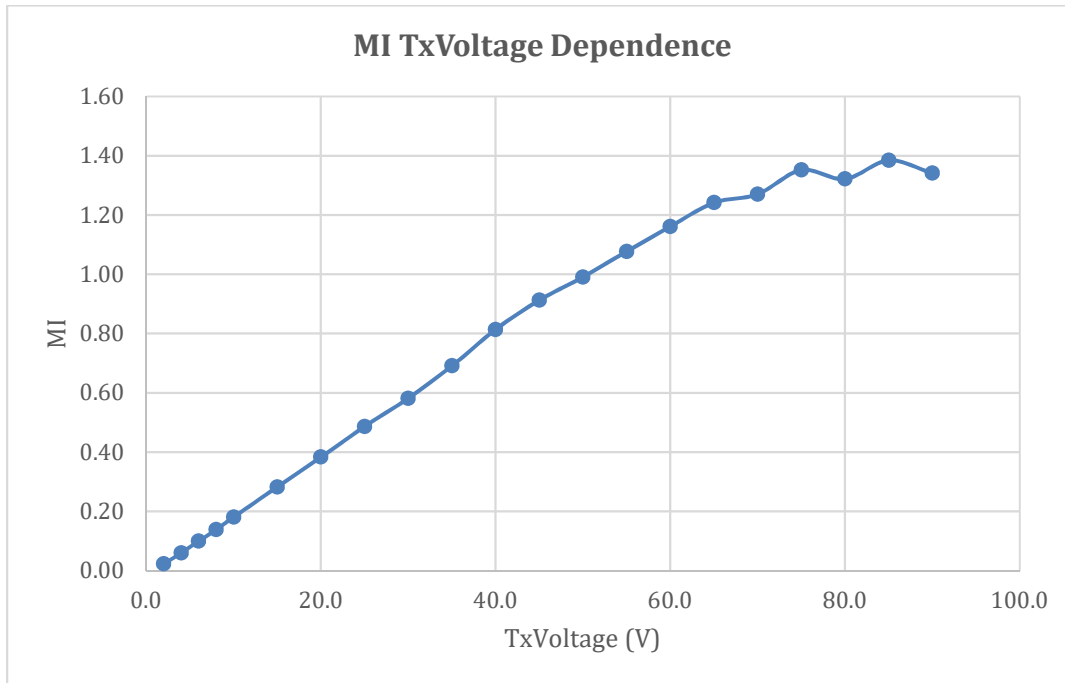


3.3 L11-5v WideBeamDoppler (D-mode)

Software release: 4.1.100-1812101200

Location: (6.50, 0.00, 9.82) mm

TxVoltage (V)	Pp (Mpa)	Pr (Mpa)	MI
2.0	0.07	0.06	0.02
4.0	0.16	0.16	0.06
6.0	0.27	0.27	0.10
8.0	0.39	0.38	0.14
10.0	0.51	0.49	0.18
15.0	0.82	0.77	0.28
20.0	1.14	1.05	0.38
25.0	1.49	1.33	0.49
30.0	1.84	1.59	0.58
35.0	2.24	1.88	0.69
40.0	2.68	2.21	0.81
45.0	3.08	2.48	0.91
50.0	3.42	2.69	0.99
55.0	3.81	2.92	1.08
60.0	4.13	3.14	1.16
65.0	4.45	3.35	1.24
70.0	4.50	3.41	1.27
75.0	4.94	3.63	1.35
80.0	4.93	3.62	1.32
85.0	5.22	3.79	1.38
90.0	4.94	3.65	1.34



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